



Introduction to cBioPortal Webinar

September 22, 2026



Webinar Agenda

- Introductory slides
- Website demo
- Chatbot demo
- MCP servers
- Q&A

Acknowledgements



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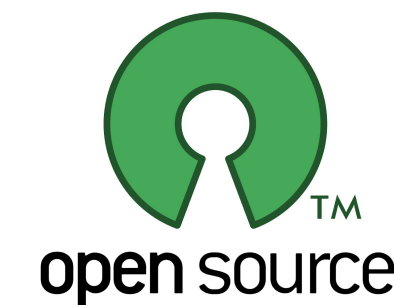
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Funding: Present & past



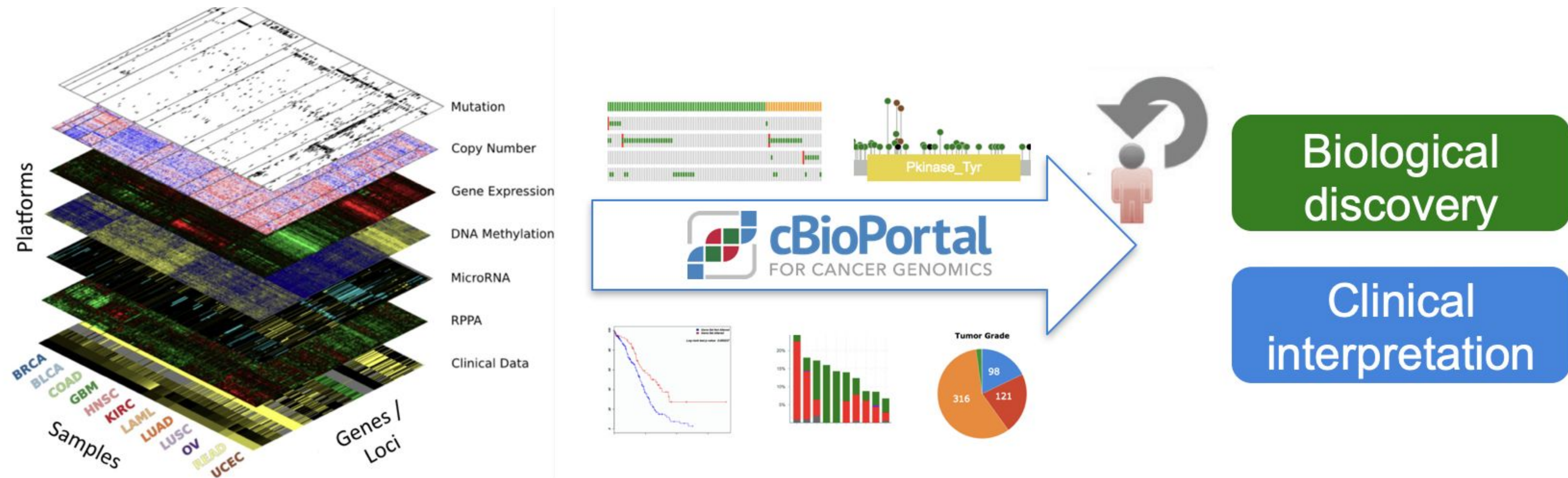
Email: cbioportal@googlegroups.com

 [@cBioPortal](https://twitter.com/cBioPortal)

 github.com/cBioPortal

cBioPortal overview

Platform for **exploratory and interactive visualization and analysis** of large-scale cancer clinico-genomics data sets



Adapted from TCGA Nat Genet 2014

Make complex cancer -omics data **accessible** and **interpretable** for cancer biologists and clinicians.

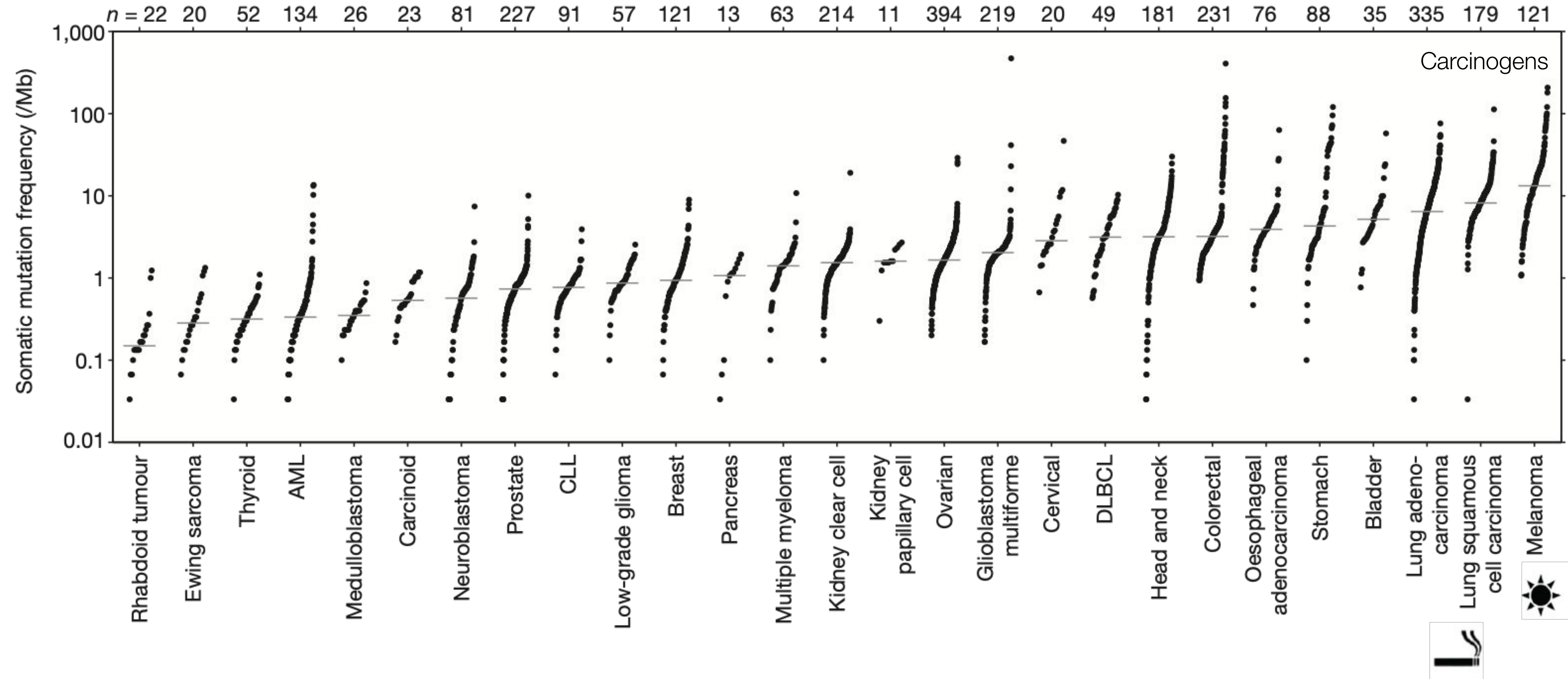
cBioPortal overview

- **Open source** software jointly developed by Dana-Farber Cancer Institute, Memorial Sloan Kettering Cancer Center, Princess Margaret Cancer Centre, Children's Hospital of Philadelphia, The Hyve and SE4Bio
- **Public website** (cBioPortal.org) with public data (TCGA, ICGC, published sequencing studies)
 - Private instances are installed at academic and commercial institutions world-wide
 - You can make OncoPrints and Lollipop plots with your own data ("Visualize Your Data" page)

cBioPortal addresses challenges in cancer -omics

- Large cancer -omics datasets are becoming increasingly common. But they can be hard to access, analyze & visualize, especially without a computational skillset
 - cBioPortal hosts data from many studies, including TCGA, for analysis and easy download
- It can be hard to interpret the data or identify the “important” events (e.g. which mutations are drivers vs passengers) in such large datasets

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- It can be hard to interpret the data or identify the “important” events (e.g. which mutations are drivers vs passengers) in such large datasets
 - cBioPortal integrates many data resources/knowledgebases

What data is in cBioPortal?

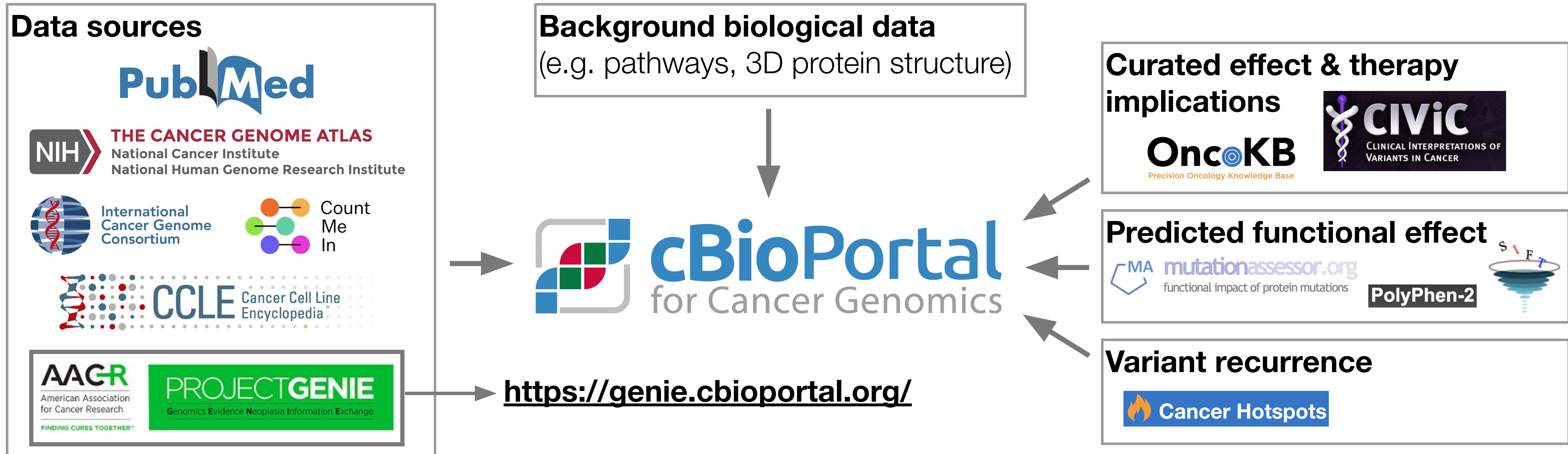
Data sources



Project:	NIH-sponsored research project	Clinical sequencing data from 19 cancer centers worldwide
Samples:	Primary, untreated tumors	Primary and metastatic tumors Pre- and post- treatment
Assays:	Whole exome sequencing + aCGH + mRNA array/seq + ...	Targeted gene panels (genes vary across cancer centers)
Data:	Mutations + copy number alterations + mRNA expression + ...	Mutations + some samples have copy number alterations

Read the original publication and understand your data source(s)

What data is in cBioPortal?



Clinical data:

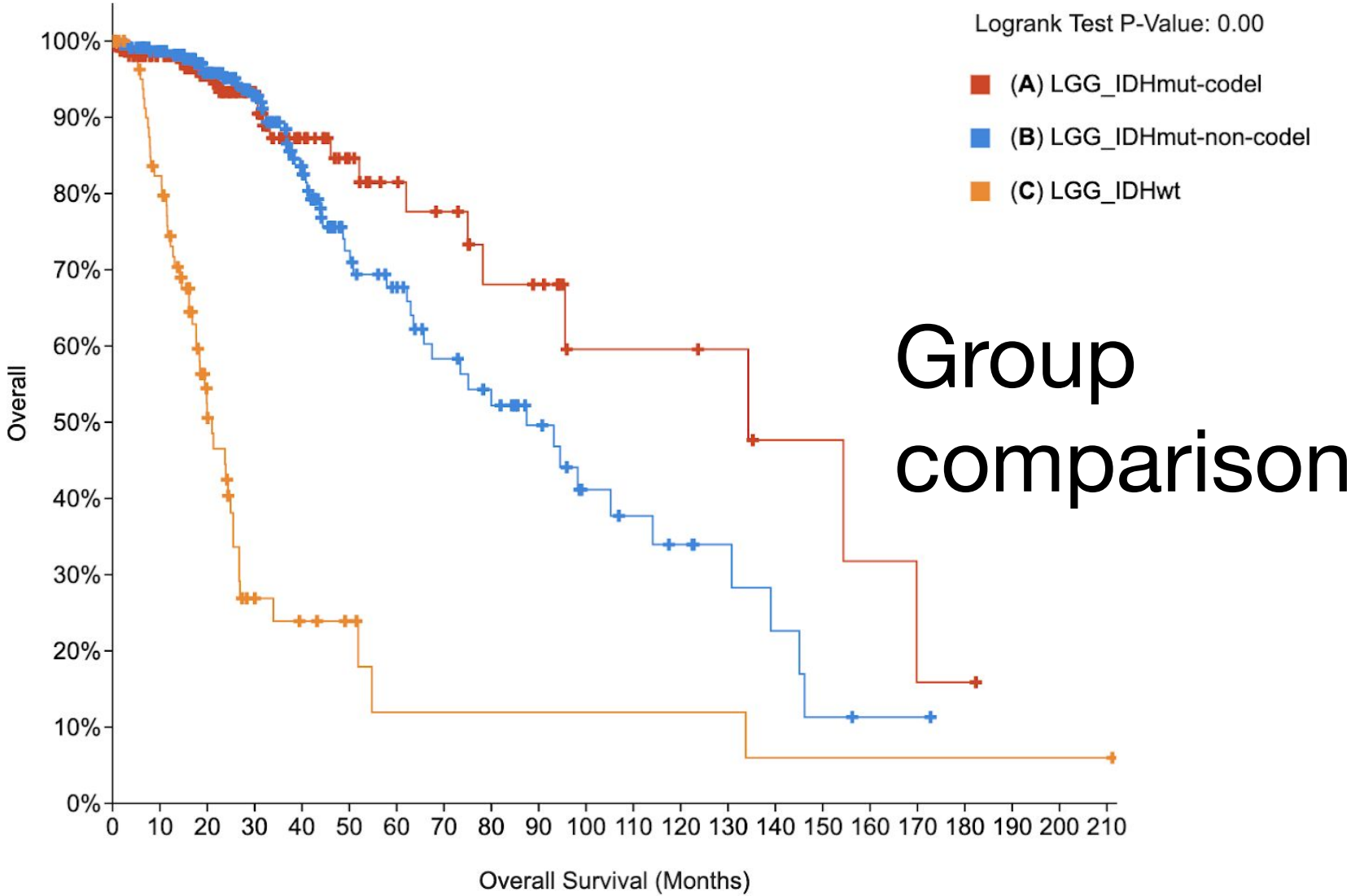
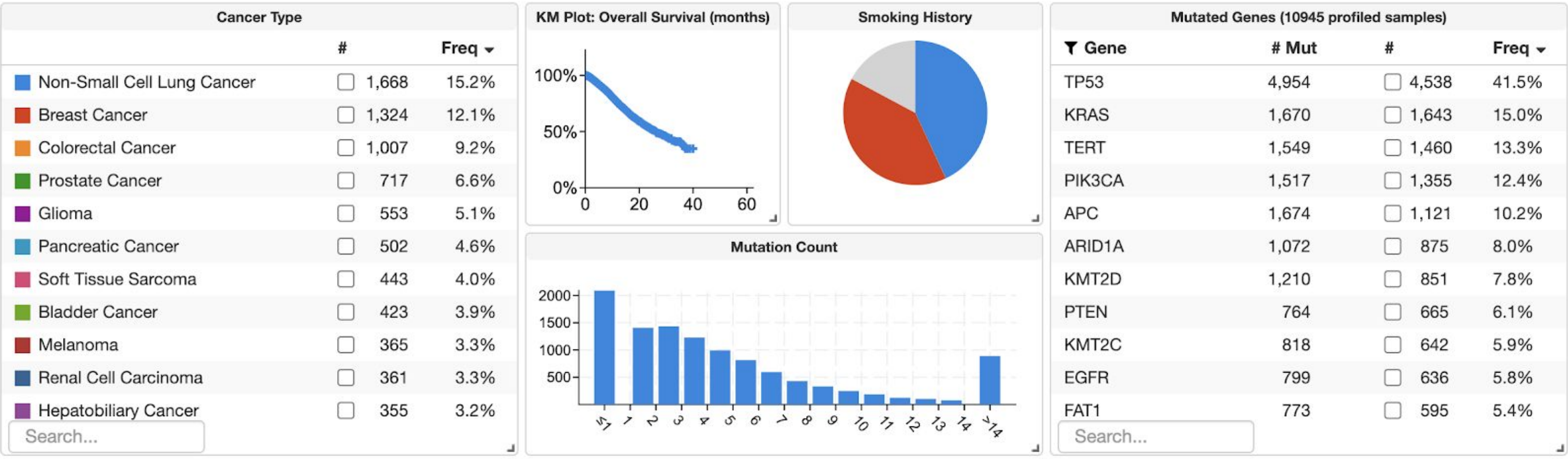
- Treatments
- Survival
- etc

-omic data:

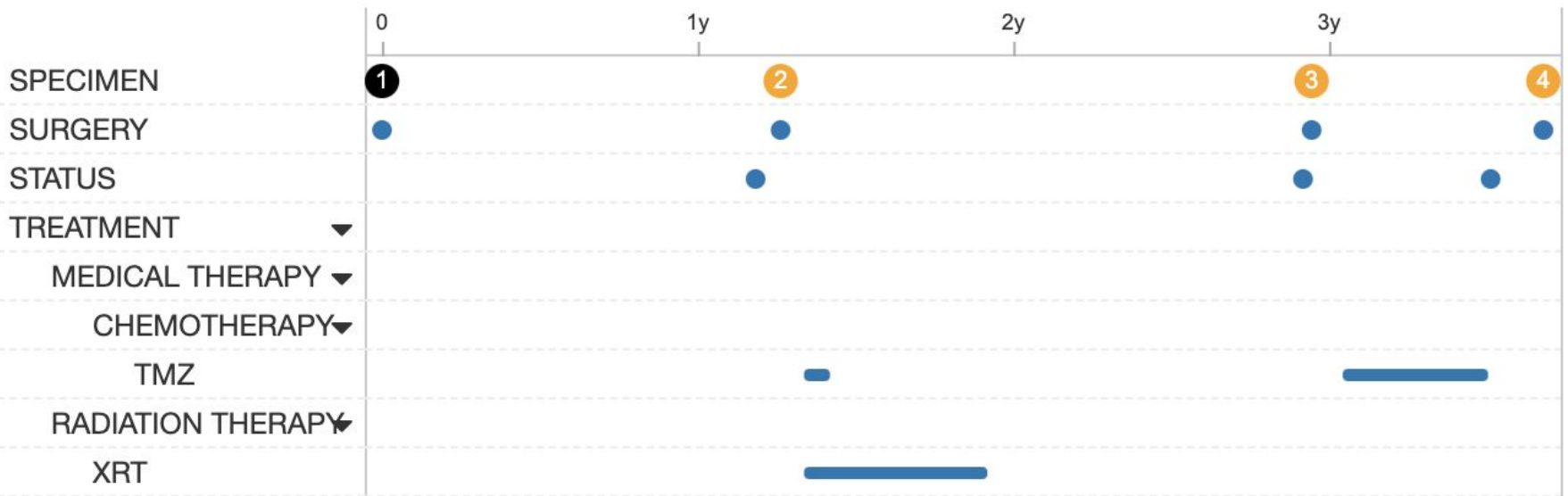
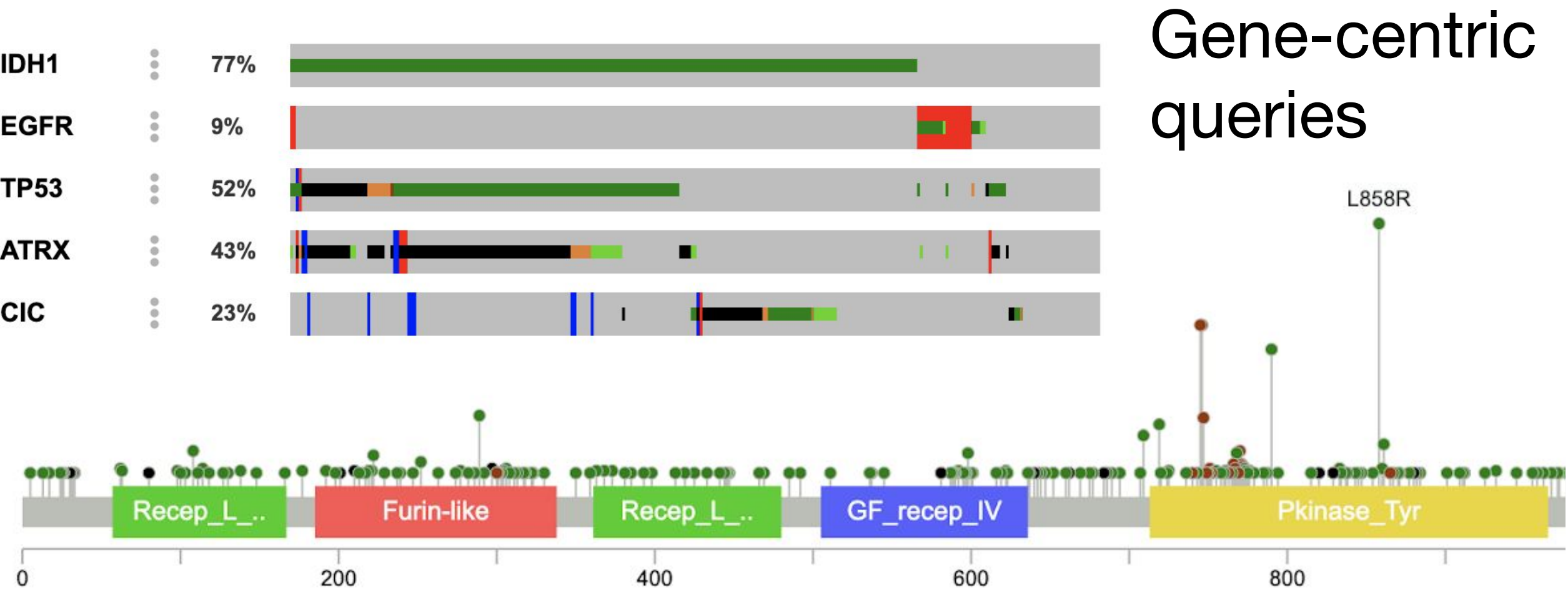
- Mutations
- Fusions
- Copy number
- mRNA expression
- Protein levels
- DNA Methylation

cBioPortal features & analysis tools

Interactive cohort exploration



Patient visualization



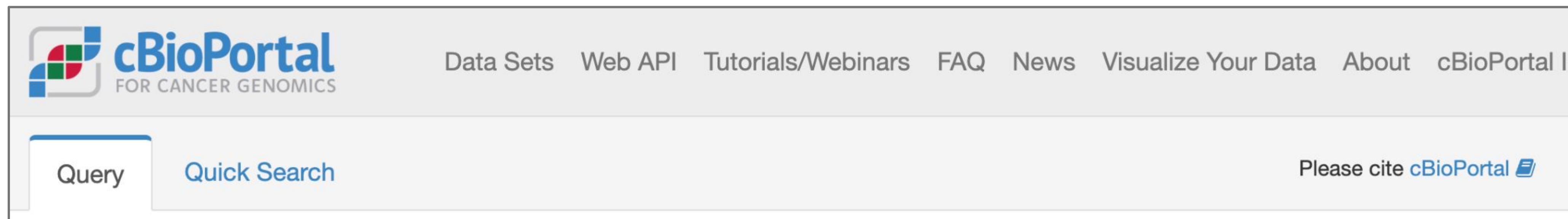
Samples	Gene	Protein Change	Annotation
1 2 3 4	IDH1	R132C	3B
1 2 3 4	TP53	R248Q	
4	ERC1	L283Fs*20	
4	HSP90AB1	K72E	
4	SPRTN	F404Lfs*3	
1	OR10V1	R273W	
1 2 3 4	TEAD3	P51L	

Getting help

- User Guide: docs.cbioportal.org
 - Step-by-step tutorial slides
 - Webinar recordings
 - Short 'how to' videos
 - FAQ
- Questions?
 - cbioportal@googlegroups.com
 - Archive: <https://groups.google.com/g/cbioportal>

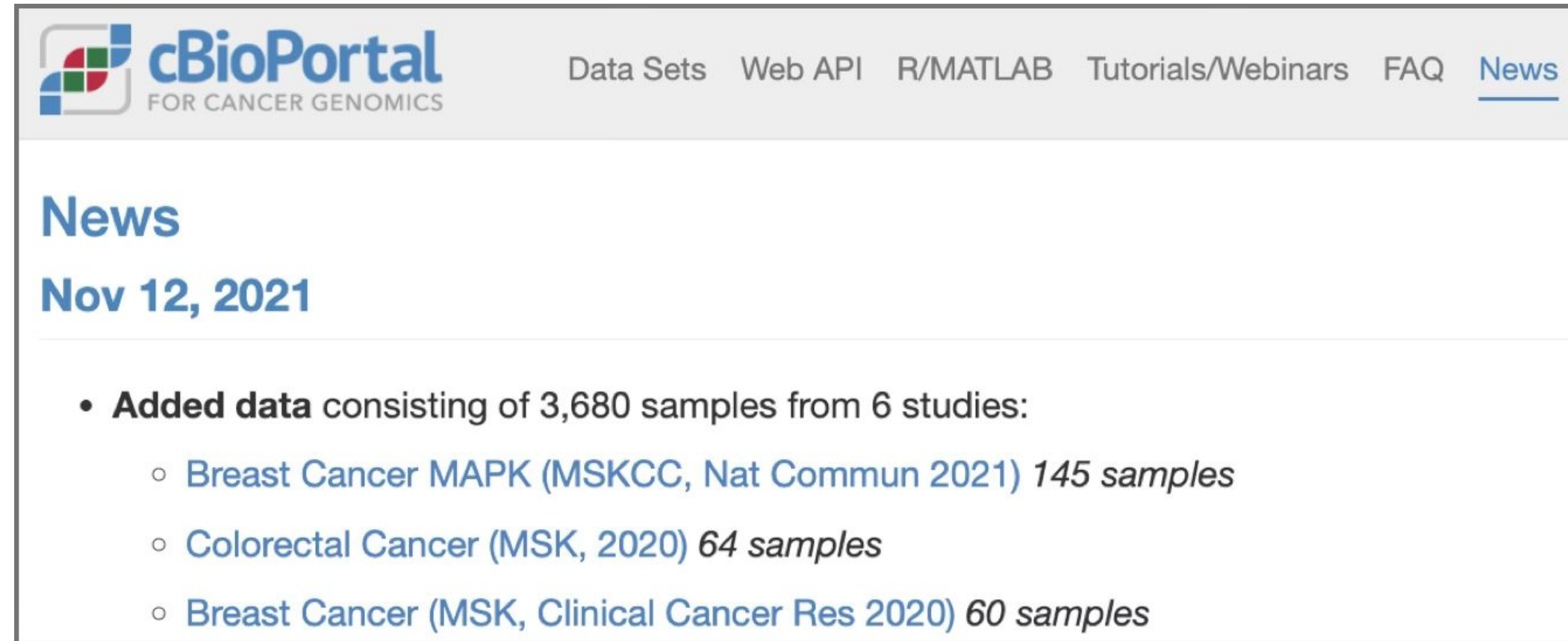
Using cBioPortal for publication

- **Please use cBioPortal in your publications & presentations!**
 - Figures are downloadable as PDF/SVG so you can customize them
- Cite the studies that generated the data you are using (if applicable)
- Cite cBioPortal



Keep up to date on cBioPortal development

News
page



The screenshot shows the cBioPortal website's News page. The header includes the cBioPortal logo and navigation links: Data Sets, Web API, R/MATLAB, Tutorials/Webinars, FAQ, and News. The main content area is titled "News" with a date of "Nov 12, 2021". It lists "Added data" consisting of 3,680 samples from 6 studies:

- **Added data** consisting of 3,680 samples from 6 studies:
 - Breast Cancer MAPK (MSKCC, Nat Commun 2021) 145 samples
 - Colorectal Cancer (MSK, 2020) 64 samples
 - Breast Cancer (MSK, Clinical Cancer Res 2020) 60 samples

Bluesky & Twitter
@cbioportal



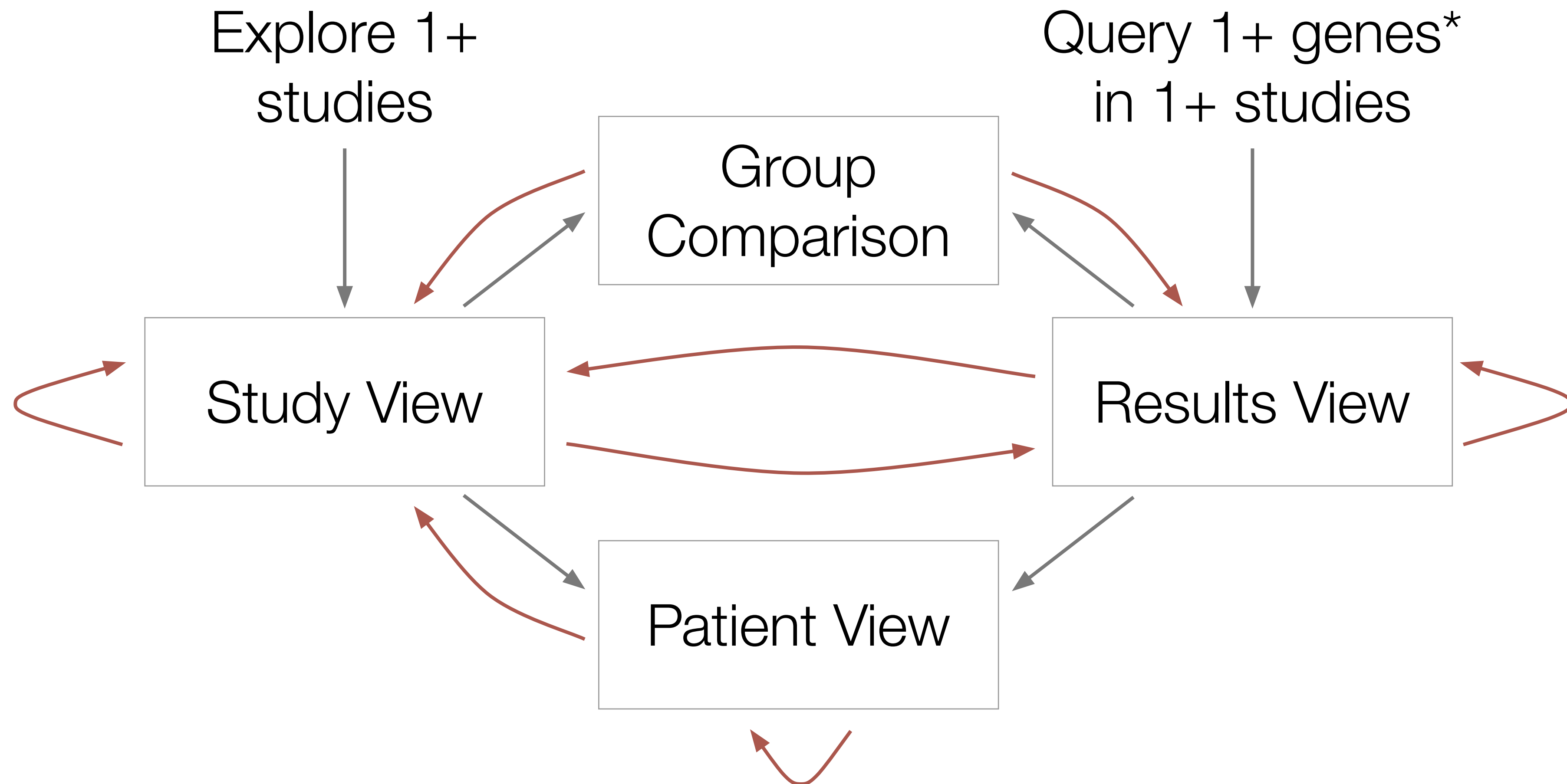
The screenshot shows a Bluesky & Twitter post from @cbioportal. The post text reads: "cBioPortal is participating in #GSOC this year! Work on Open Source Software for a summer and support Cancer Research. Read more: github.com/cbioportal/gsoc". Below the text is a promotional graphic for the Google Summer of Code (GSOC) featuring the cBioPortal logo and the text: "Work on Open Source Software for a summer and support Cancer Research! Paid internship - any age any background! Flexible schedule (175h or 350h projects) Read more: github.com/cbioportal/gsoc". The graphic also includes the Google Summer of Code logo and the text "Applications due April 8th 18:00 UTC".



The screenshot shows the cBioPortal Newsletter. The header features the cBioPortal logo and the text "NEWSLETTER". The main content area is titled "Welcome to the December 2024 edition of the cBioPortal Newsletter!". Below the title is the cBioPortal logo and the text "1,918 followers". The date "December 20, 2024" is displayed. The main text reads: "As the year comes to a close, we're thrilled to highlight the latest advancements and improvements in cBioPortal. From new features to enriched datasets, there's plenty to explore. Join us as we take a closer look at what's new and exciting!"

Newsletter
Subscribe via LinkedIn or
Google Groups

cBioPortal workflow



* Can customize alteration types & thresholds per gene

<https://www.cbioportal.org/>

cBioChat

Ask questions: “***Tell me about PTEN mutation in LUAD***”

Get answers in two ways:

1: Links to cBioPortal pages
(navigation)

2: Direct answers from cBioPortal
data (database)



Ask cBioPortalChat to explore cBioPortal data, build direct cBioPortal links, and answer questions about studies, genes, mutations, clinical attributes, and patient-level results.

[Learn more](#)

How can I help you explore cBioPortal?



🕒 20 msgs / 30 min · 📄 20M tokens / week

[Hide examples](#)



Explore Data

Find studies, datasets, and available molecular data in cBioPortal.



Navigate cBioPortal

Create direct links to cBioPortal views, cohorts, plots, and study pages.



Analyze Data

Compare genes, cancer types, molecular profiles, cohorts, and outcomes.



Please provide feedback!  

<https://chat.cbioportal.org/>

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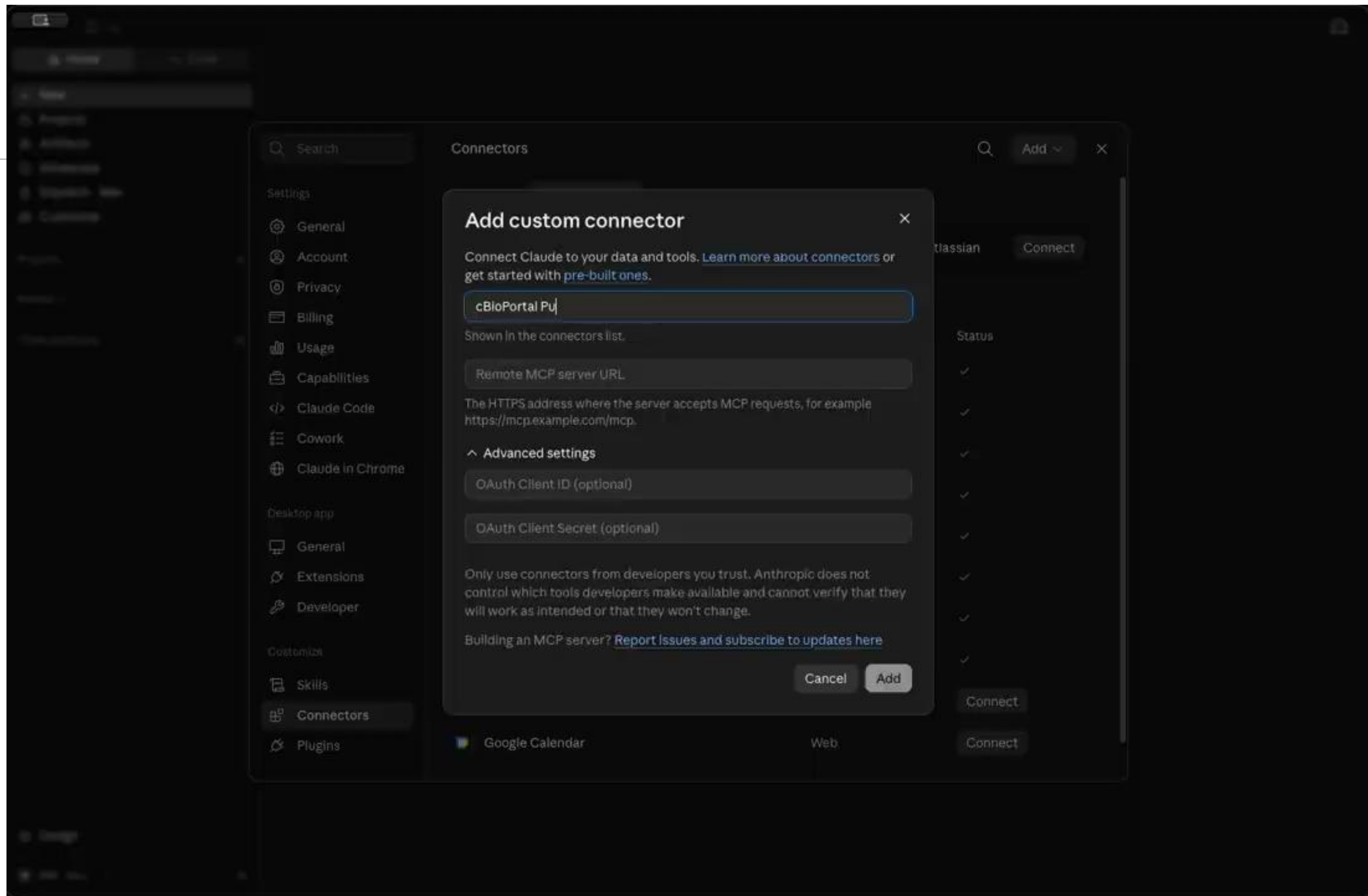
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MCP servers

1: <https://mcp.cbioportal.org/navigator/mcp>

2: <https://mcp.cbioportal.org/db/mcp>



Q&A Session

- A recording of this webinar will be posted shortly
- Please fill out the post-webinar survey

Thank you for attending!